

X20 System

User's Manual

Version: **1.20 (June 2006)**

Model No.: **MAX20-ENG**

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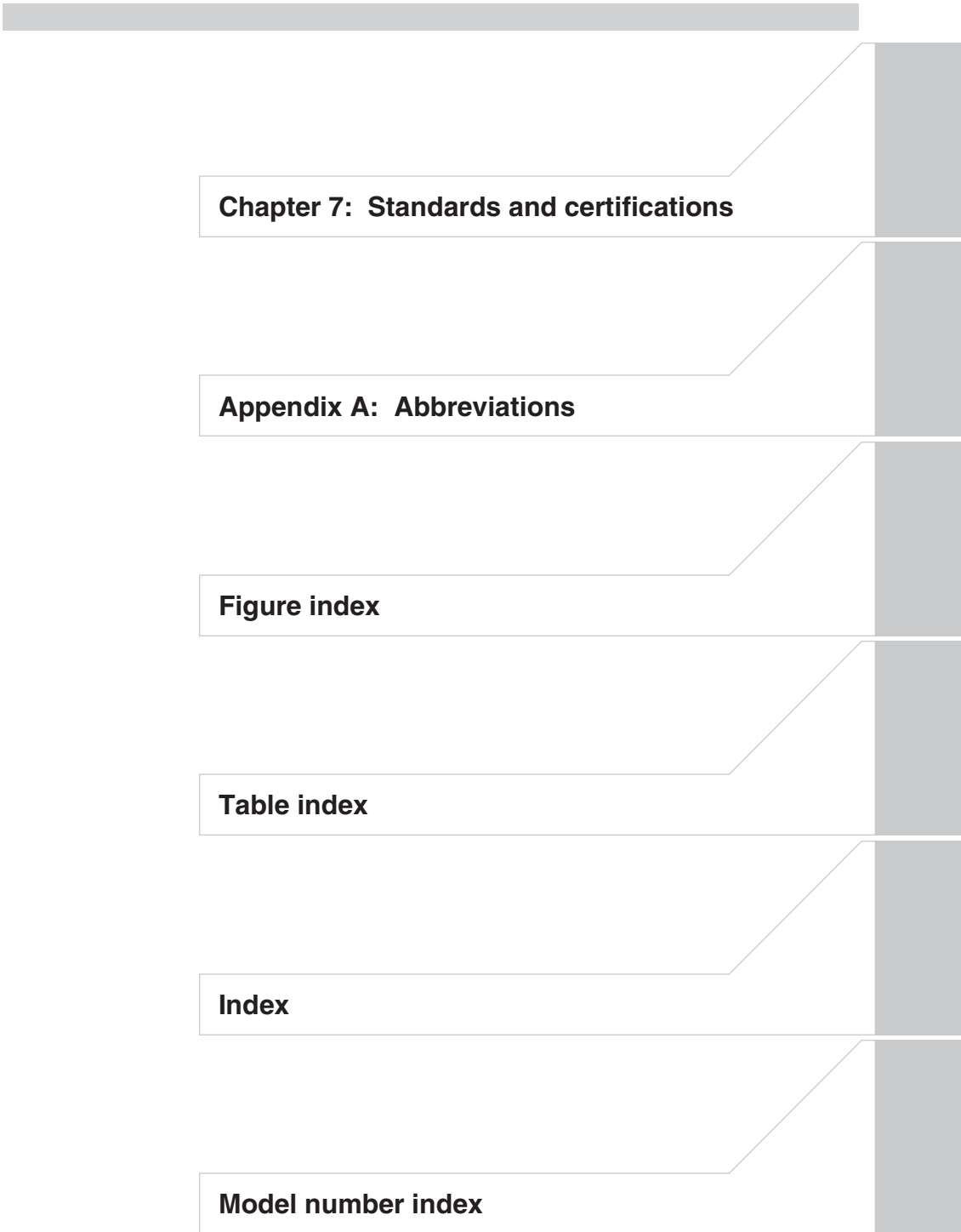
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Chapter 1 • General information

1. Manual history

Version	Date	Comment
1.20	June 2006	First edition

Table 1: Manual history

2. Safety notices

2.1 Introduction

Programmable logic controllers (PLCs), operating and monitoring devices (industrial PCs, Power Panels, Mobile Panels, etc.), as well as the B&R uninterruptible power supplies have been designed, developed or manufactured for conventional use in industry. They were not designed, developed and manufactured for any use involving serious risks or hazards that could lead to death, injury, serious physical damage, or loss of any kind without the implementation of exceptionally stringent safety precautions. In particular, such risks and hazards include the use of these devices to monitor nuclear reactions in nuclear power plants, as well as flight control systems, flight safety, the control of mass transportation systems, medical life support systems, and the control of weapons systems.

When using both programmable logic controllers and operating and monitoring devices as control systems in conjunction with a Soft PLC (e. g. B&R Automation Runtime or comparable products) or a Slot PLC (e. g. B&R LS251 or comparable products), the safety precautions applying to industrial control systems (e. g. the provision of safety devices such as emergency stop circuits, etc.) in accordance with applicable national and international regulations must be observed. The same applies for all other devices connected to the system, e. g. drives.

All tasks such as installation, commissioning, and service may only be carried out by qualified personnel. Qualified personnel are persons familiar with transport, mounting, installation, commissioning and operation of the product and have the respective qualifications (e.g. IEC 60364). National accident prevention guidelines must be followed.

The safety guidelines, connection descriptions (rating plate and documentation) and limit values listed in the technical data must be read carefully and must be observed before installation and commissioning.

2.2 Intended use

Electronic devices are generally not failsafe. In the event of a failure on the programmable control system, operating or monitoring device or uninterruptible power supply, the user is responsible for ensuring that other devices that may be connected, such as motors, are made secure.

2.3 Protection against electrostatic discharges

Electrical components that are vulnerable to electrostatic discharge (ESD) must be handled accordingly.

2.3.1 Packaging

- Electrical components with housing
... do not require special ESD packaging, but must be handled properly (see section "Electrical components with housing").
- Electrical components without housing
... must be protected by ESD-suitable packaging.

2.3.2 Guidelines for proper ESD handling

Electrical components with housing

- Do not touch the contacts of connectors on connected cables.
- Do not touch the contact tips on the circuit boards.

Electrical components without housing

In addition to "Electrical components with housing", the following also applies:

- Any persons handling electrical components or devices that will be installed in the electrical components must be grounded.
- Components can only be touched on the small sides or on the front plate.
- Components should always be stored in a suitable medium (ESD packaging, conductive foam, etc.).
Metallic surfaces are not suitable storage surfaces!
- Electrostatic discharges should be avoided on the components (e.g. through charged plastics).
- A minimum distance of 10cm must be kept from monitors and TV sets.
- Measurement devices and equipment must be grounded.
- Measurement probes on potential-free measurement devices must be discharged on sufficiently grounded surfaces before taking measurements.

Individual components

- ESD protective measures for individual components are thoroughly integrated at B&R (conductive floors, footwear, arm bands, etc.).
- The increased ESD protective measures for individual components are not necessary for our customers for handling B&R products.

2.4 Transport and storage

During transport and storage, devices must be protected from excessive stress (mechanical load, temperature, humidity, aggressive atmosphere, etc.).

The devices contain components sensitive to electrostatic charges which can be damaged by inappropriate handling. It is therefore necessary to provide the required safety precautions against electrostatic discharges during installation or removal of the devices (see also section 2.3 "Protection against electrostatic discharges" on page 27).

2.5 Installation

- Installation must take place according to the documentation using suitable equipment and tools.
- Devices may only be installed without voltage applied and by qualified personnel.
- General safety regulations and nationally applicable accident prevention guidelines must be observed.
- Electrical installation must be carried out according to the relevant guidelines (e.g. line cross section, fuse, protective ground connection).
- Take the necessary steps to protect against electrostatic discharges (see also section 2.3 "Protection against electrostatic discharges" on page 27).

2.6 Operation

2.6.1 Protection against touching electrical parts

To operate programmable logic controllers, operating and monitoring devices or uninterruptible power supplies, certain components must carry dangerous voltage levels of over 42 VDC. A life-threatening electrical shock could occur if you come into contact with these parts. This could result in death, severe injury, or material damage.

Before turning on the programmable logic controller, the operational and monitoring devices and the uninterruptible power supply, ensure that the housing is properly grounded (PE rail). The ground connection must be established when testing the operating and monitoring devices or the uninterruptible power supply, even when operating them for only a short time.

Before turning the device on, make sure that all voltage-carrying parts are securely covered. During operation, all covers must remain closed.

2.7 Organization of safety notices

The safety notices in this manual are organized as follows:

Safety notice	Description
Danger!	Disregarding the safety regulations and guidelines can be life-threatening.
Caution!	Disregarding the safety regulations and guidelines can result in severe injury or major damage to material.
Warning!	Disregarding the safety regulations and guidelines can result in injury or damage to material.
Information:	Important information for preventing errors.

Table 2: Organization of safety notices

3. Definition of terms

Term	Explanation
SG3	System Generation 3 (SG3) - CPUs with Motorola processors. The following CPUs belong to this series: CP260, IF161, IP161, XP152, CP100, CP104, CP152, CP153, CP200, CP210, CP430, CP470, CP474, CP476, CP770, CP774, PP21, PP41
SG4	System Generation 4 (SG4) - CPUs with Intel processors. The following CPUs belong to this series: PP100, CP340, CP360, CP380, CP382, AR102, AR105, AR010, AR000

Table 3: Definition of terms

Chapter 2 • System characteristics

1. The new standard for automation

There are many different I/O slice systems. According to the motto "Perfection in Automation", B&R is setting new standards with the X20 System. Born from experience gained from applications all over the world, numerous conversations with customers, and with the aim for more simple, economical and secure usage, the X20 System is the new universal solution for any automation task in machine and system manufacturing.

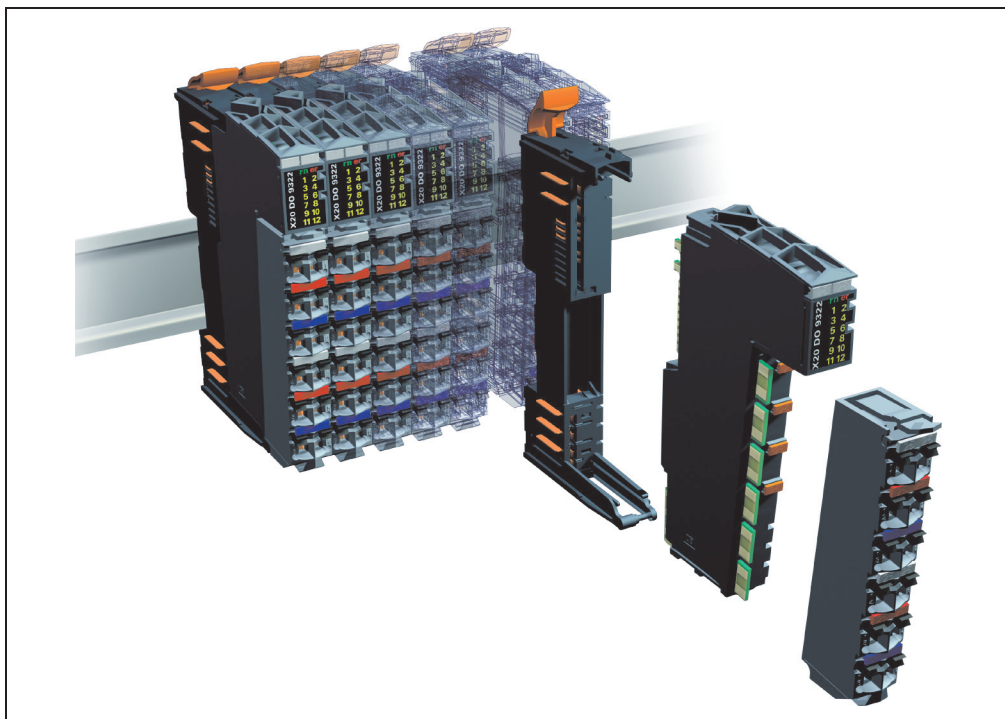


Figure 1: Three basic elements result in one module: Terminal block – Electronic module – Bus module

1.1 More than just I/O

With well thought-out details and a sophisticated ergonomic design, the X20 System is more than a remote I/O system, it is a complete control solution. The X20 System family makes it possible to combine the exact components necessary to adapt to the user's demands and individual application requirements.

- The X20 System is the ideal addition to a standard fieldbus and expands the possibilities of standard control systems. Simply connect it, configure it, and you're done.
- Teamed up with other B&R components, the X20 System achieves its full potential and allows the implementation of applications with unimagined performance and flexibility. Seamless integration is a major advantage.

1.2 3 x 1 = One

Three basic elements result in one module: Terminal block – Electronic module – Bus module

This modularity results in a system that combines the advantages of both rack and I/O slice systems:

- Prewiring without the module
- Hot pluggable electronics
- Extra bus slots for added options

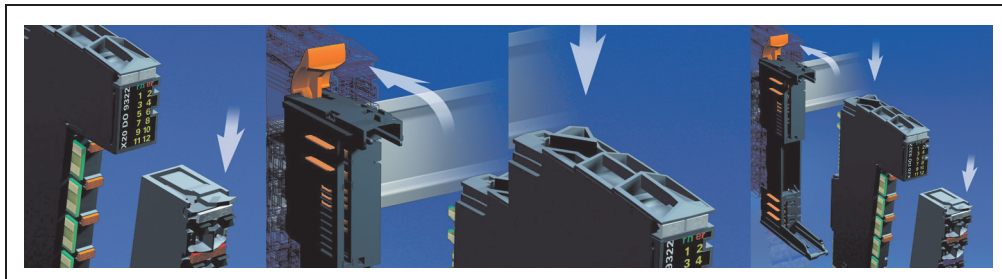


Figure 2: X20 modules are divided into three parts to guarantee the simplest usability

The X20 System is distinguished by a 50% increase in component density, perfected connection technology and optimal granularity.

- **Added value**

12 channels with a width of 12.5 mm allow an unprecedented component density, with optimal terminal ergonomics. As a result, the X20 system offers 50 % more channels than conventional slice systems. And this without sacrificing terminal connections.

- **Continuity**

Consistent implementation of 1-wire, 2-wire or 3-wire connections - no additional jumper terminals needed.

- **Granularity**

One channel and two channel modules: Maximum flexibility so you only have to pay for what you really need.

2. Optimally designed

X20 modules are divided into three parts to guarantee the simplest usability throughout their entire lifecycle. The division into bus module, electronic module and terminal block provides many advantages.

- **Preconfigured for different machine types**

The X20 System bus modules are the basic platform for many machine variations. The machine's intended purpose determines which electronics modules are used. The software recognizes the design automatically and provides the necessary functions. Handling a range of machine types couldn't be easier.

- **Industrial switching cabinet construction**

The X20 System terminal blocks, which are separated from the electronics modules, make it possible to prewire complete switching cabinets. Ideal for series production machines.

- **Easy maintenance**

X20 modules can be easily exchanged to simplify troubleshooting. The electronics modules are exchanged without interrupting operation. The wiring stays the same thanks to the separate terminal blocks. Being able to exchange the automation components quickly reduces down-time.

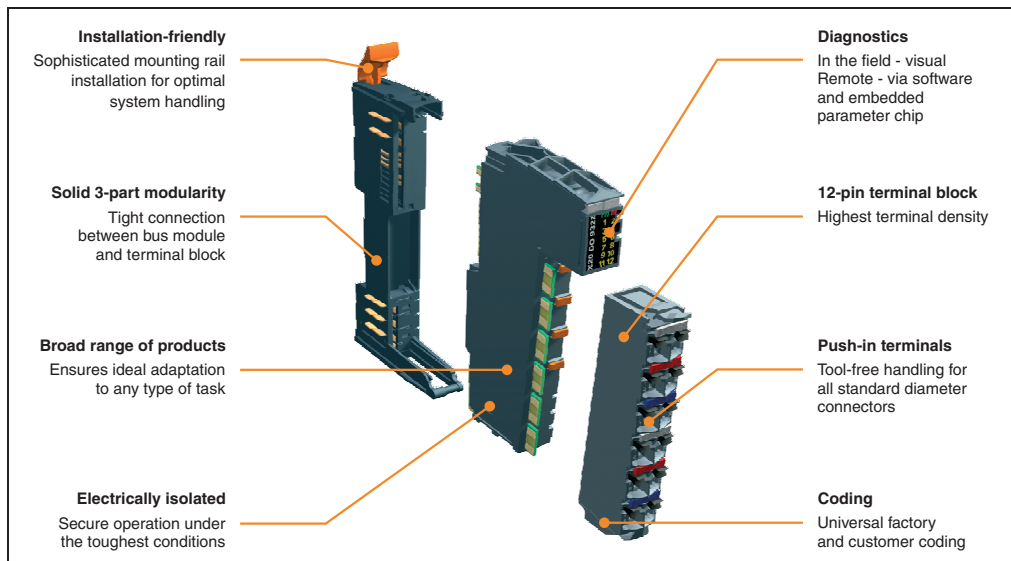


Figure 3: Features of the X20 System

3. Decentralized backplane

The main idea: a decentralized backplane for a rack system - in other words, the cable is the backplane. All modules are connected to the uniform backplane (X2X Link). Directly connected X20, X67 or XV modules can each be placed at a distance of up to 100 m outside the confines of the switching cabinet. X2X Link guarantees resistance to disturbances at the highest possible level based on twisted copper cables.

This not only provides a universal remote backplane which handles the communication between bus modules and via the X2X Link cable, but makes it possible without converters or any loss in performance. A unique feature of the X20 is the possibility to later integrate machine options on bus modules that are not yet being used without having to change the software addressing.

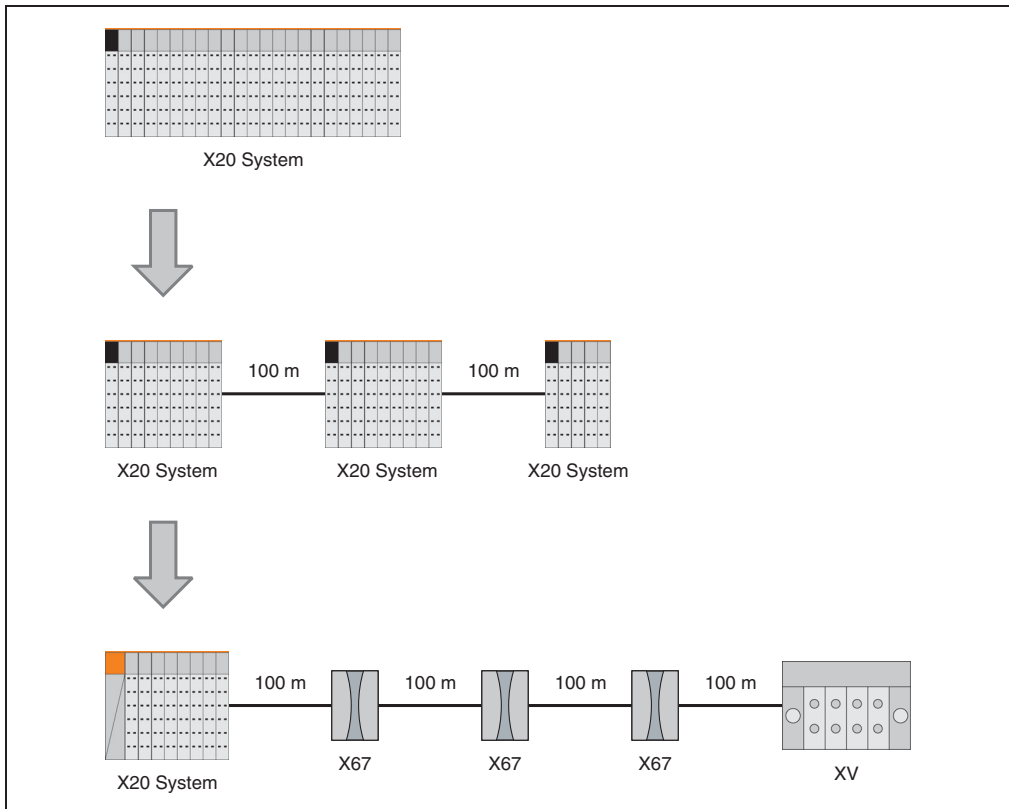


Figure 4: X2X Link - universal backplane based on twisted copper cables

4. For all fieldbuses, integrated through the standard

The X20 System is ideally suited for expanding existing control systems using standard fieldbus technology.

Using bus controllers, the X20 System can be used as a powerful I/O expansion unit. Standardized EDS or GSD description files allow X20 System components to be integrated, configured, and programmed as usual in the programming environment of a non-B&R system.

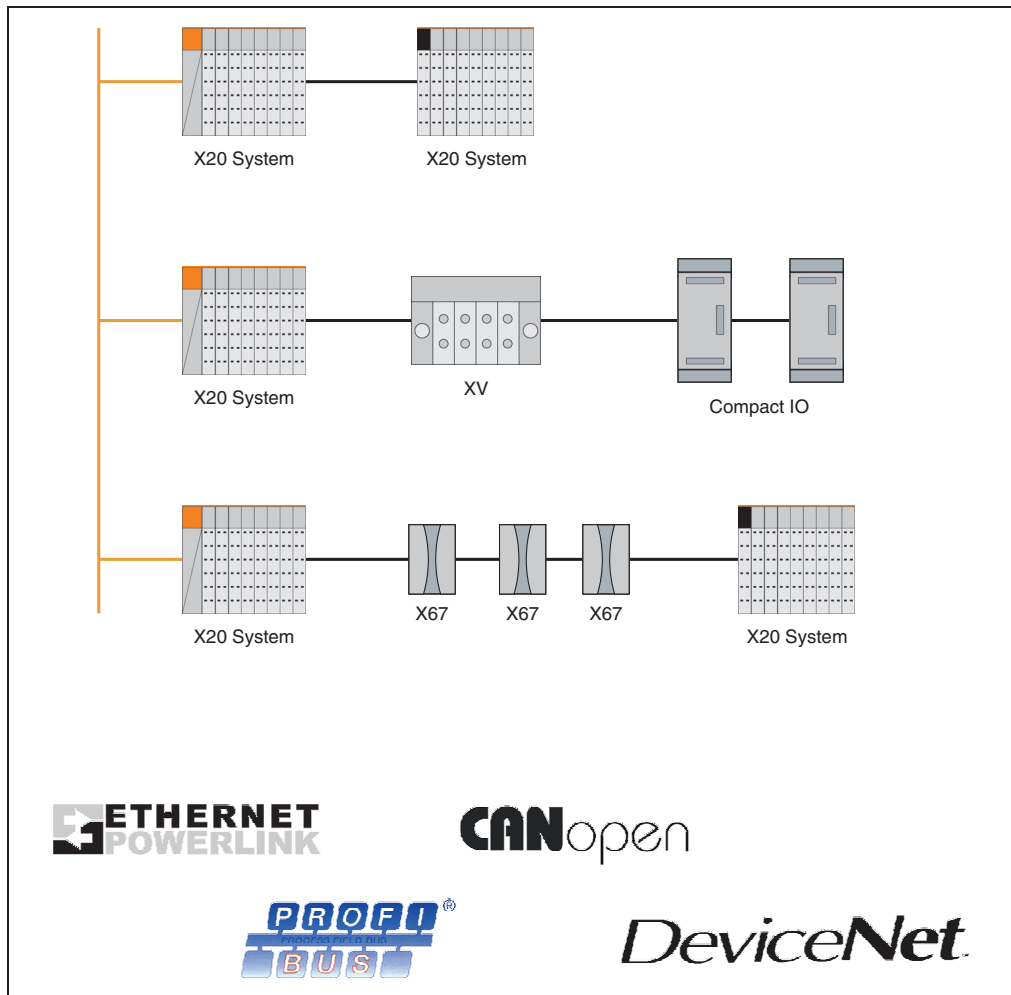


Figure 5: Expansion of existing control systems using standard fieldbuses and the X20 System

5. Complete system

5.1 IP67 - then X67

The X67 is the robust version of the X20 for use outside the switching cabinet. The same basic technology, with an extremely robust housing and modules with between 4 and 32 channels, guarantees economical solutions in the roughest conditions.

5.2 Integrated valve manifold control

The development of the XV System allows for the first time direct and manufacturer-independent control of valve manifolds. A complete digital output module in a size and form comparable with a normal DSUB connector. XV allows any valve manifold manufacturer to be selected because it is connected directly to the standardized multipulse connector on the valve manifold.

Fully integrated in the remote backplane, it rounds off the X20 and X67 for complete automation solutions. One system, several variations - advantages that pay off. You select your automation components and distribute them as needed inside and outside the switching cabinet.

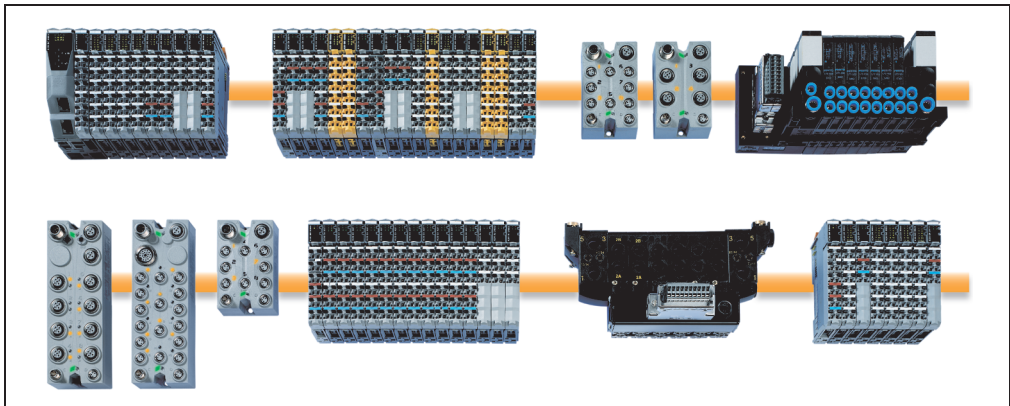


Figure 6: X20, X67, XV - variations of a single system

6. Easy wiring

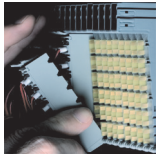
Industrial switching cabinet construction streamlines production cycles. Prefabricated cable trees enable faster and easier assembly directly on the machine or system. The X20 System supports efficient prewiring of the entire switching cabinet using separate terminal blocks. The complete X20 System configuration is mounted in the switching cabinet and connected to the prewired cable trees.

The supply of the X20 modules and the supply of the sensors and actuators do not add any requirements for energy distribution. The X20 System reduces manual wiring to a minimum.

6.1 Install the wires, plug it in, and it's ready to go

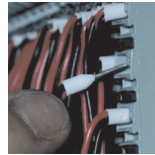
Simple, tool-free wiring for fast installation. The X20 System terminal blocks use a fully-integrated and proven push-in connector system. Each terminal can also handle double wire sleeves up to a diameter of $2 \times 0.75 \text{ mm}^2$. The user saves time wiring the system and distributing the signals.

The wire connections can be removed with a screwdriver. Each terminal also has an access point for a measurement probe. A great deal of thought was given to designing every aspect the X20 System. Right down to the wire connectors.



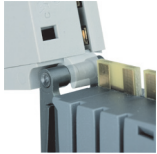
Detached

The terminals can be prewired apart from the actual I/O module. This provides many advantages for switching cabinet construction. Separate manufacturing, just-in-time logistics and the installation of preassembled systems during start-up become reality.



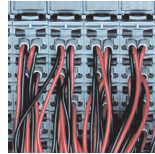
Tool-free

Simple, tool-free wiring for fast installation. The X20 System terminals use a fully-integrated and proven push-in connector system. Available with 6-pin and extremely compact 12-pin terminals.



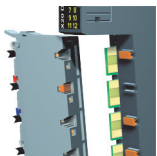
Coded in the system

Factory coding prevents dangerous mix-ups. The coding guarantees that only parts which are permitted to be combined can be combined. Intuitively and without additional work.



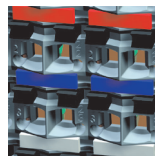
Ergonomic

Component density must not come at the expense of ergonomics. With terminal spacing of more than 5 mm, this was handled optimally on the X20 System. Experience gained in the field - used in the field.



Coded in the application

Incorrectly inserting terminals does not necessarily damage the electronics, but always causes faulty functioning of the system. Application coding prevents this problem.



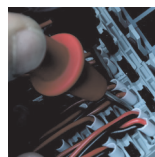
Clarity

Distinct forms intuitively define various functions, such as clearly assigned latching and unlatching functions for terminals. This prevents errors from the very beginning.



Labeling

Each terminal is clearly labeled, directly in the plastic. Additional label tabs are available as system accessories including a printer with ECAD connection.



Easy servicing

A system's strengths can be seen in its details: In addition to the terminal connector and unlocking mechanism, each terminal has an access point for a test probe. You can easily measure the terminal potential without disconnecting the wire.

7. Sophisticated mechanics

The name B&R stands for many years of experience in developing and manufacturing industrial electronics. But it's also the mechanics of the X20 System that have been thought through to the last detail. Its robust design, long guides, and strengthened housing guarantee the stability it needs in industrial environments. These features allow the X20 System to be mounted on a rail with the same ease as a rack system. They also make it just as simple to remove it from the rail.

The sophisticated mechanics of the X20 are needed not just to provide this type of handling, but also to be able to quickly and easily remove I/O slices from the entire system.

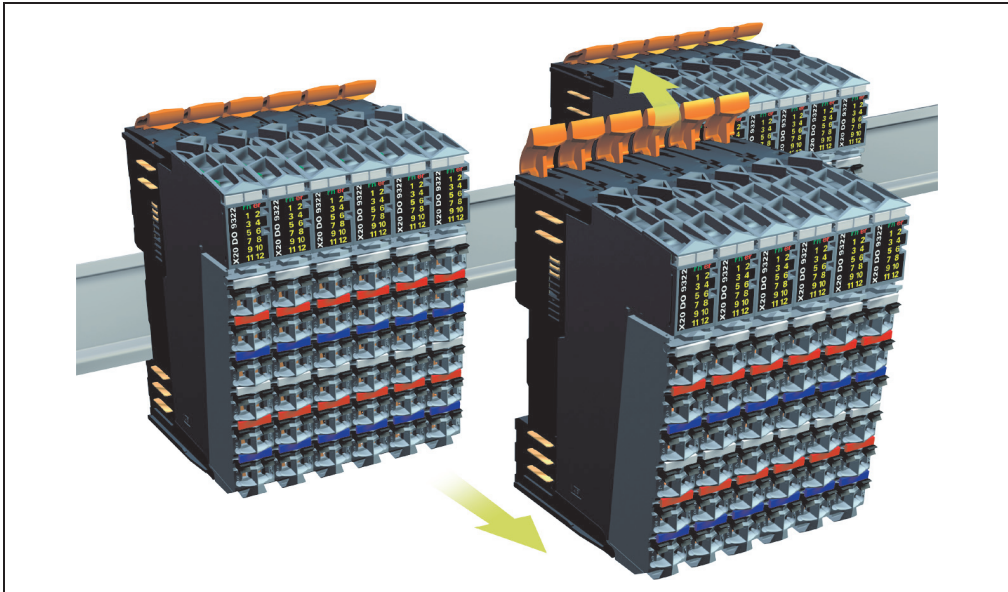
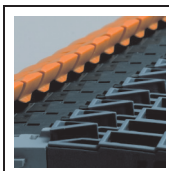
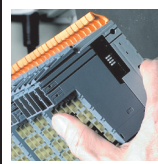


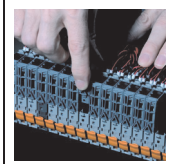
Figure 7: Easy mounting on and removal from the mounting rail



Unlocking lever with two positions
Closed for secure fit on the mounting rail.



Clear open position makes the difference
Open to remove a module or the entire system.



Remove a single module from the system
Remove or re-insert vertically.



Mount the entire system as a whole
Or just as easily remove the the entire system.

8. Diagnostics

Only with outstanding diagnostic options can errors be found quickly. The X20 System offers several levels of diagnostics:

- Directly on the module using visual LED displays. Bus status, I/O status and channel states are displayed in direct assignment to the channels or the function. The different states are displayed differently, e.g. green for OK, red for error.
- Via software in the cyclical data image. With the X20 System, status data means no additional communication effort, which would result in considerable differences between theoretically possible bus speeds and real requirements during operation. All necessary status data is always transferred cyclically, with no exceptions.
- Expanded diagnostic data in non-cyclic data traffic without loss in performance. If a problem occurs, detailed diagnostic data can be requested from the application by the respective module using an asynchronous channel. There is no additional communication load whatsoever, and cycle times remain unchanged.

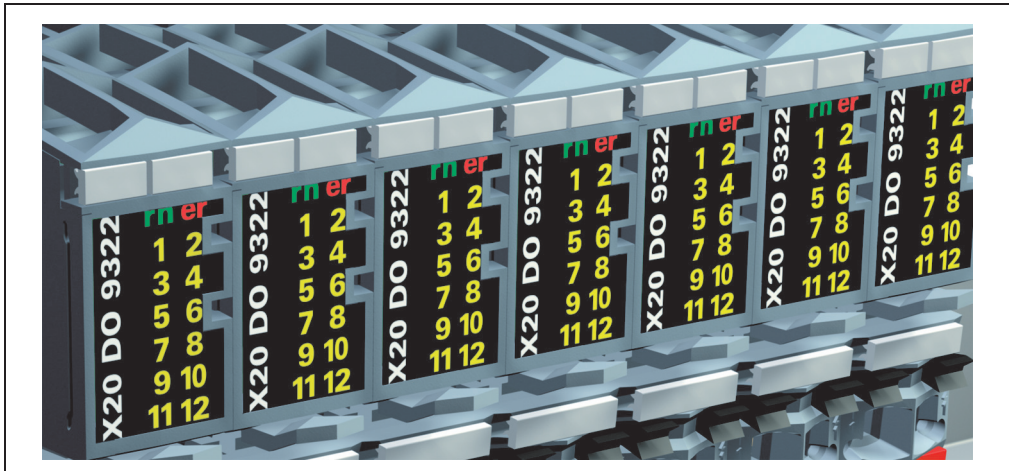


Figure 8: Visual diagnostics directly on the module using LED displays

9. Embedded parameter chip

Information such as module type, serial number, functionality and version number is contained in the embedded parameter chip of the X20 module. This information is automatically evaluated by the Automation Studio™ programming environment and by the application program. This prevents errors, both with commissioning and with service. In addition, the system configuration is automated and flexible variations are made possible.

Worldwide defined serial numbers of modules are gaining increasing significance in validated systems as demanded, for example, in the FDA.

10. Space for options

Depending on the user's demands and individual application requirements, the X20 System family makes it possible to combine the exact components necessary. This allows machine options to be implemented easily and flexibly. Bus modules provide the basis. They are more or less a rack replacement. Depending on the option, only the necessary electronic modules are then inserted in the predefined slots.

Addresses are assigned implicitly using the slot. Software that has been developed once is valid for all versions and does not need to be changed. This even applies to later machine expansion. The I/O modules are simply inserted in the defined bus modules, and assigned the corresponding potential groups and E-stop groups.

To prevent unwanted expansion, each module can be identified and then enabled using the application software.

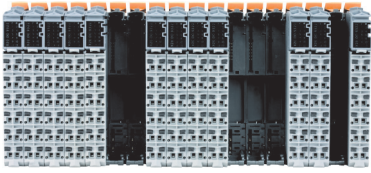
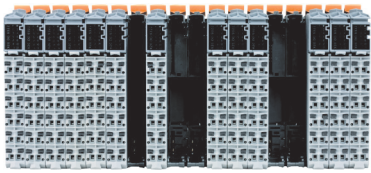
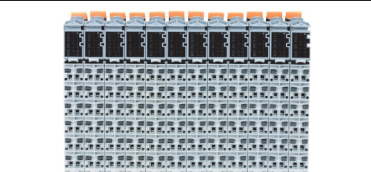
11. Flexibility for options

The implementation of different machine variations using free bus modules is only one of the many features that the X20 System offers. With the support of B&R Automation Studio™, there is an optimized solution using I/O configuration. What does this mean?

Each I/O configuration is created optimally according to the actual need. However, the application software is already designed for all options. Only the I/O channels that are actually available are configured for the application program. If an expansion is required, then the additional hardware needed can be easily connected and the configuration of the I/Os changed. This is done without having to compile the application software.

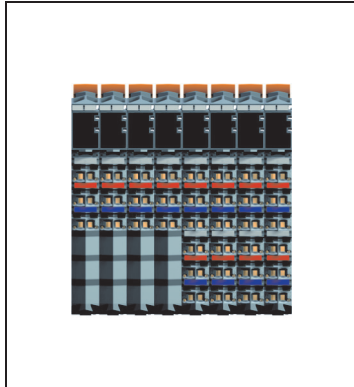
It doesn't matter how the I/O configuration list is created:

- Manually in B&R Automation Studio™
- With tools, e.g. with a database or a table calculation program
- Directly from an ERP system, just like the parts list for the machine
- Automatically in the application software, depending on the hardware used

	Machine variation A The possibilities of the X20 System can be explained using examples. This is a machine constellation with two variations, A and B. All of the necessary electronics modules for machine variation A are shown in the picture to the left. The bus modules needed for variation B are also present, but without electronic modules.
	Machine variation B Variation B shows the necessary electronic modules but the modules necessary for variation A are missing. The distribution of the free bus modules for the variations is also clear: The variable I/O modules can be very easily connected to the necessary electrically isolated groups and don't need to be attached in the back. The involved process of taking apart the configuration to expand existing electrically isolated groups is also eliminated. Simply insert the electronic module and put on the terminal block.
	Machine variation A - optimized The features included in Automation Studio™ can also be used to achieve completely optimized hardware configuration without losing the advantage of comprehensive application software for all variations. As described earlier, simply switching physical I/O points to the application program makes it extremely easy to optimize the hardware variations without even needing compilation.

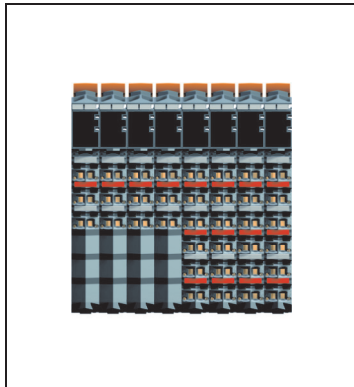
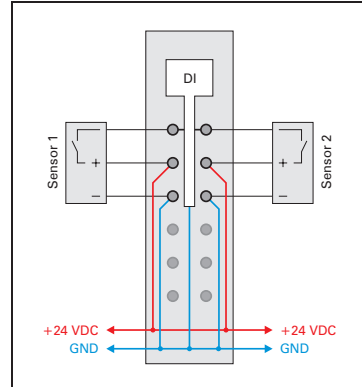
12. Universal 1-line, 2-line, and 3-line connections

Consistent connection types for all requirements – no additional jumper terminals are needed. All connection types can also be mixed and matched.



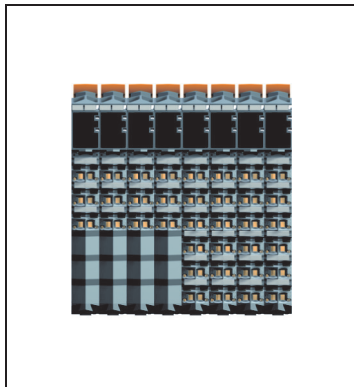
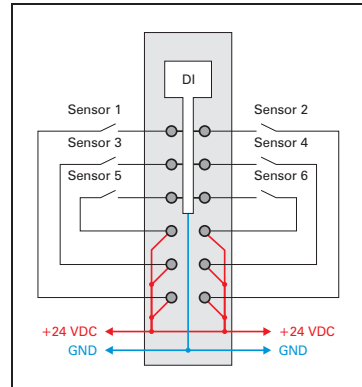
Universal 3-wire connections

Integrated supply and ground for sensors and actuators.



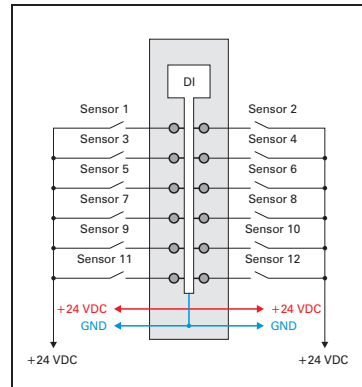
Universal 2-wire connections

Extra terminals are not needed.



Universal 1-wire connections

12 channels - unequalled component density.



13. X20 System configuration

The X20 System is designed so that can be connected to either standard fieldbuses (with a bus controller) or the remote X2X Link backplane (with a bus receiver). The connection to the next station is made with a bus transmitter. Supply modules and I/O modules are placed between the bus receiver or bus controller and the bus transmitter as needed.

The power supply system used in the X20 Systems is described in chapter 3 "Mechanical and electrical configuration", section 6 "The power supply design" on page 52.

13.1 Fieldbus connection

Several bus controllers for standard fieldbus technologies like ETHERNET Powerlink, Device Net, Profibus or CANopen are available to connect X20 modules to existing control systems. Fieldbus configurations transparently integrate the X20 System into the 3rd-party development environment.

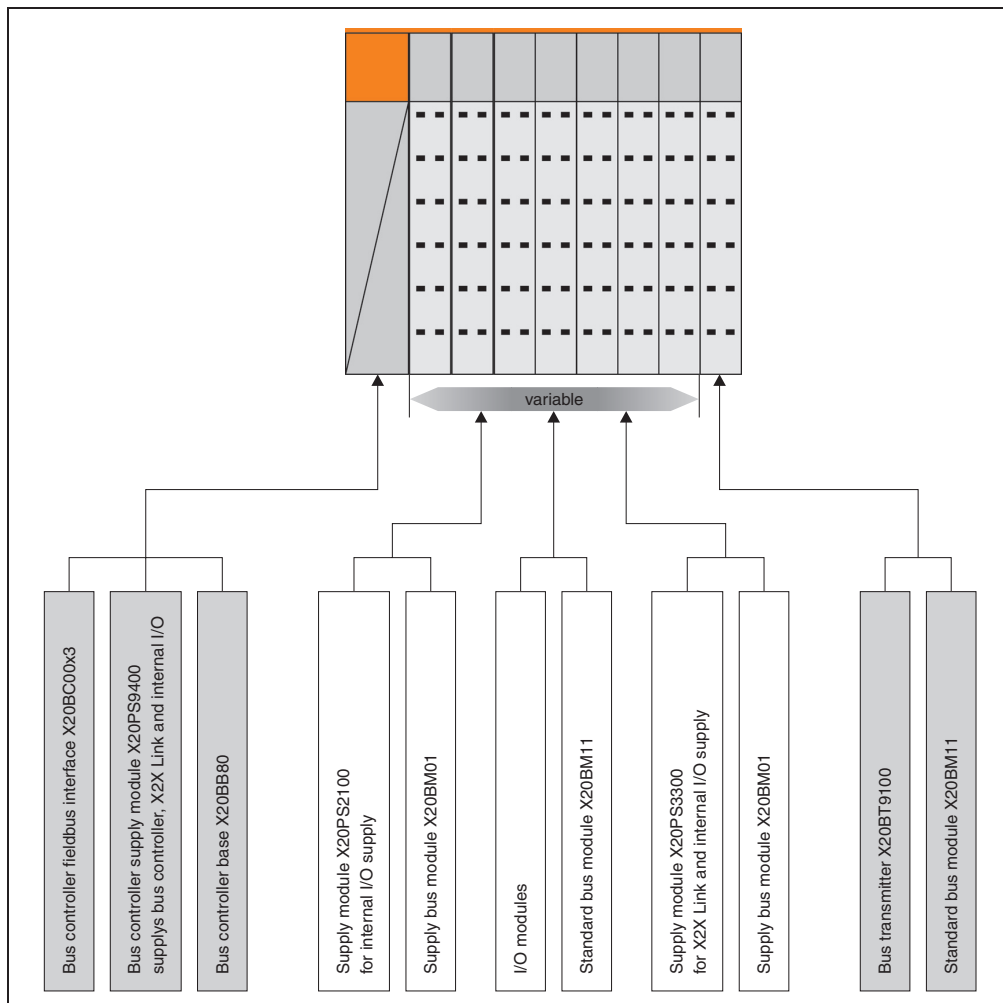


Figure 9: X20 System configurator for fieldbus connection

13.2 Connection to X2X Link backplane

The bus receiver BR9300 is used to connect the X20 System directly to the remote X2X link backplane.

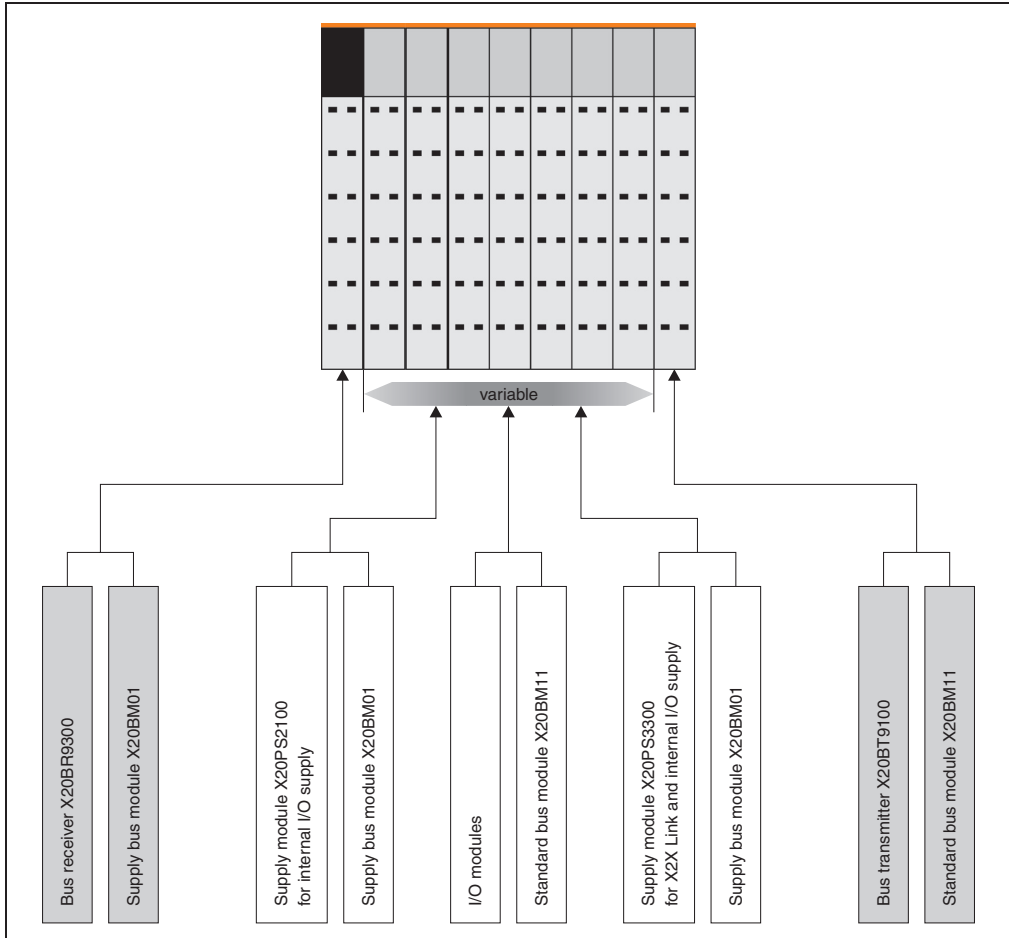


Figure 10: X20 System configurator for connection to X2X Link backplane

Chapter 3 • Mechanical and electrical configuration

1. Dimensions

1.1 I/O modules

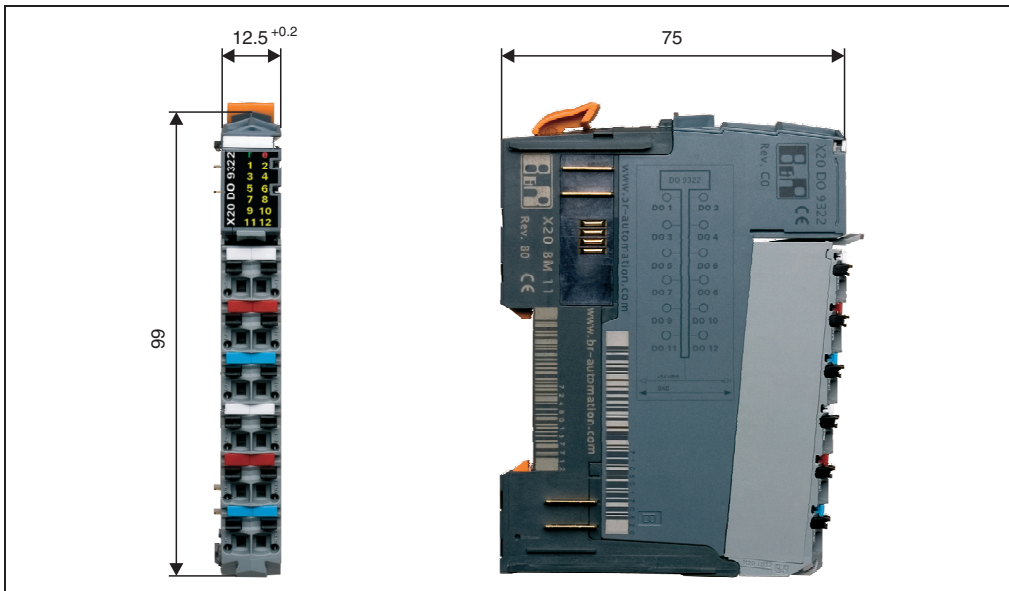


Figure 11: X20 System - I/O module dimensions

1.2 Bus controller

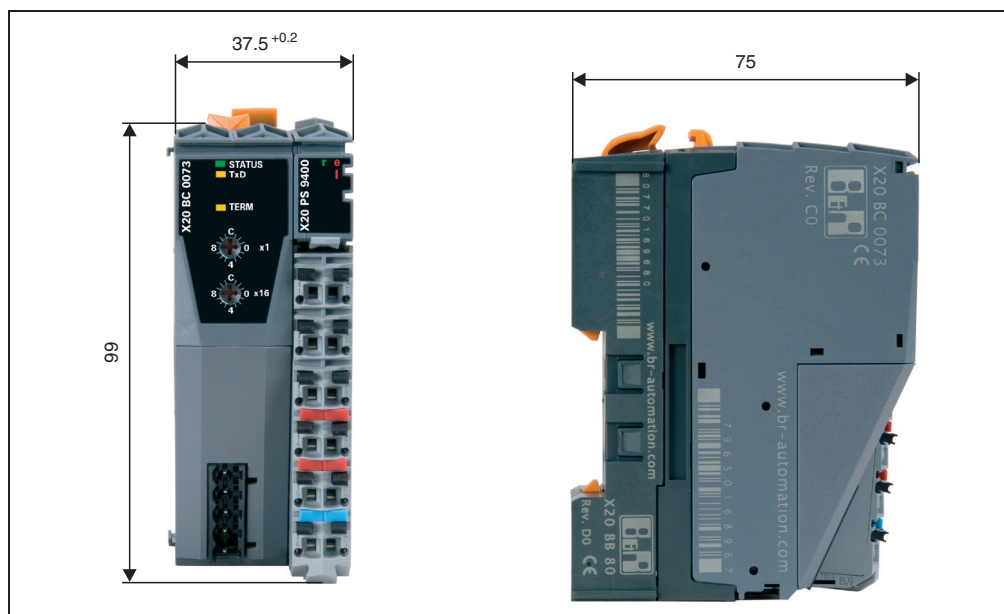


Figure 12: X20 System - bus controller dimensions

1.3 CAD support

To ensure CAD support, the dimensions are in 2D for the ECAD macros. STEP data is provided for 3D representation.

The STEP data can be downloaded from the B&R website www.br-automation.com under Services.

2. Construction support

2.1 Macros for ECAD systems

The electronics in a machine must be designed in a manner which optimally utilizes the materials and space available. Graphic ECAD systems have established themselves as the right tool for the job.

Every module in the X20 System comes with pre-designed electronic descriptions of the mechanical dimensions, electrical signals and module functions. These macros are loaded directly to the well-established ECAD systems. The wiring plans are automatically applied by the configuration and programming system, Automation Studio™. Design and changes are immediately reflected at all levels of development. This saves time for the more important tasks and prevents errors right from the start. The accelerated development, programming, maintenance and documentation involved with the X20 System mean lower costs, enhanced quality and increased sales by earlier entry into the market.

2.2 Printing support

System printers and standard identification labels are supported by the appropriate printer software. Printing manually, from table calculations, or directly from ECAD software – all methods are supported. The software and printer systems are from the company Weidmüller.

3. Installation guidelines

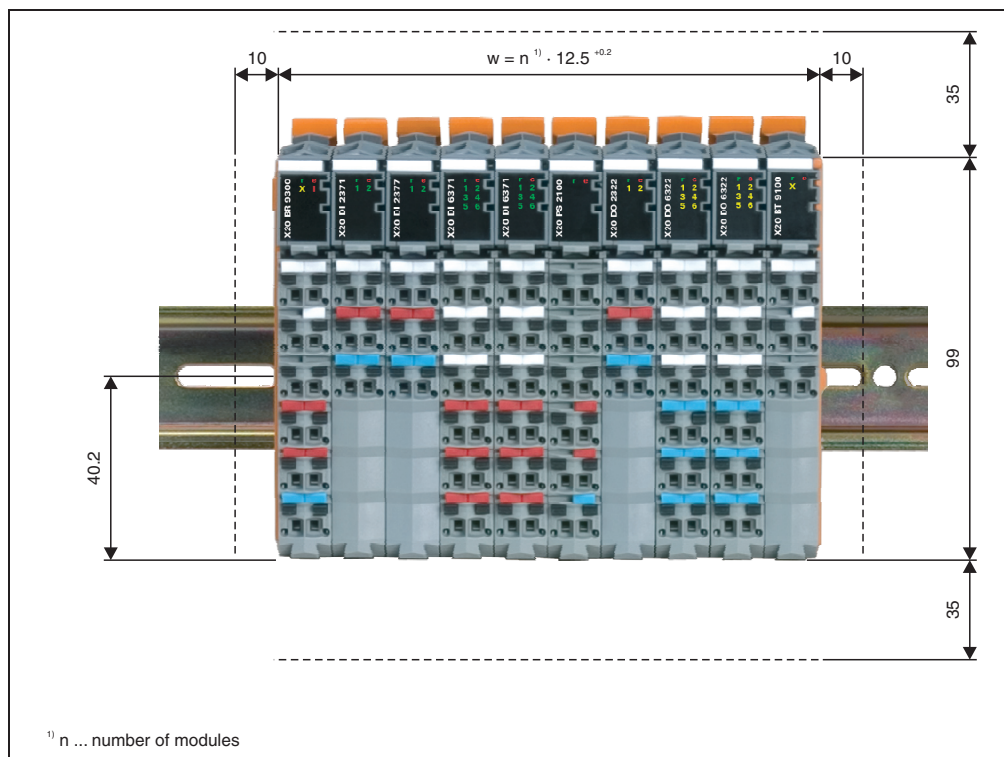


Figure 13: X20 System - installation guidelines

For optimal cooling and air circulation, there must be at least 35 mm free space above the modules. To the left and right of the X20 System, there must be at least 10 mm of free space. Underneath the module, 35 mm space must be left free for the input, output and supply cables.

4. Installation

A mounting rail conforming to the EN 60715 standard (TH35-7.5) is required to mount the PLC. The mounting rail is fastened to the back wall of the switching cabinet.

The entire system of individual modules is hung in the desired location on the mounting rail with the locking levers open and locked in place by closing the levers. Finally, the modules are equipped with the prewired terminal blocks.

5. Stress relief using cable ties

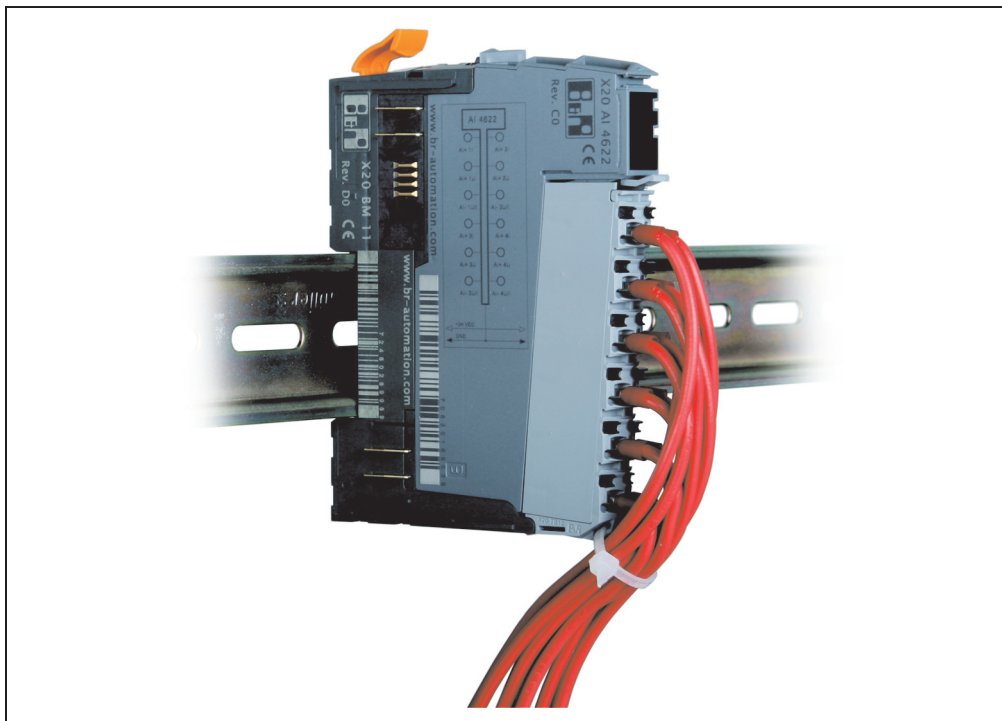


Figure 14: Stress relief using cable ties

The X20 System terminal blocks have slots for the cable ties. If needed, a cable tie can be fed through these slots to reduce the stress on the cable.

Cable tie dimensions: width ≤ 4.0 mm
 thickness ≤ 1.2 mm

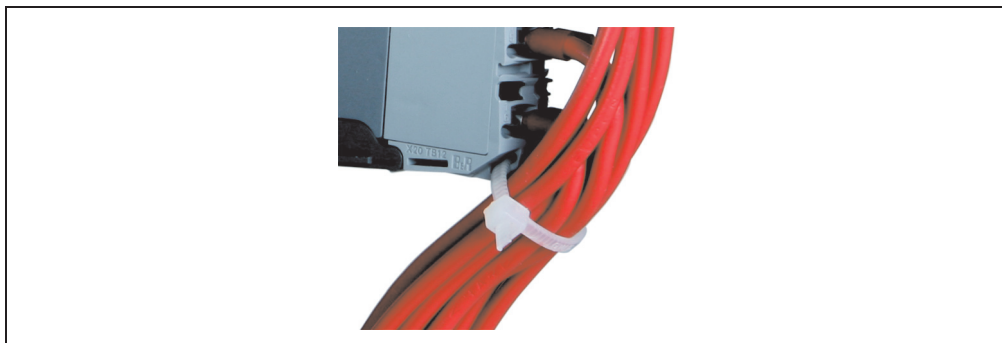


Figure 15: Slots through which the cable ties are fed

6. The power supply design

6.1 Bus module replaces rack

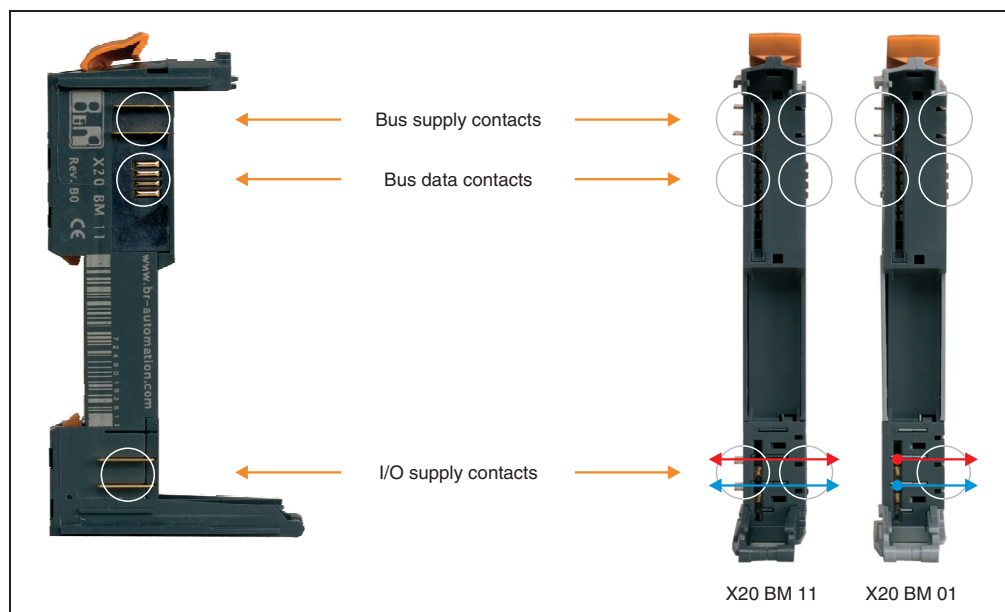


Figure 16: The bus module replaces the rack in the X20 System

The bus module is the backbone of the X20 System regarding the bus supply and bus data as well as the I/O supply for the electronic modules. Each bus module is an active bus station, even without electronic module. There are two variations of the bus module:

- Interconnected I/O supply
- Supply module with the I/O supply split to the left

7. X20 System infrastructure

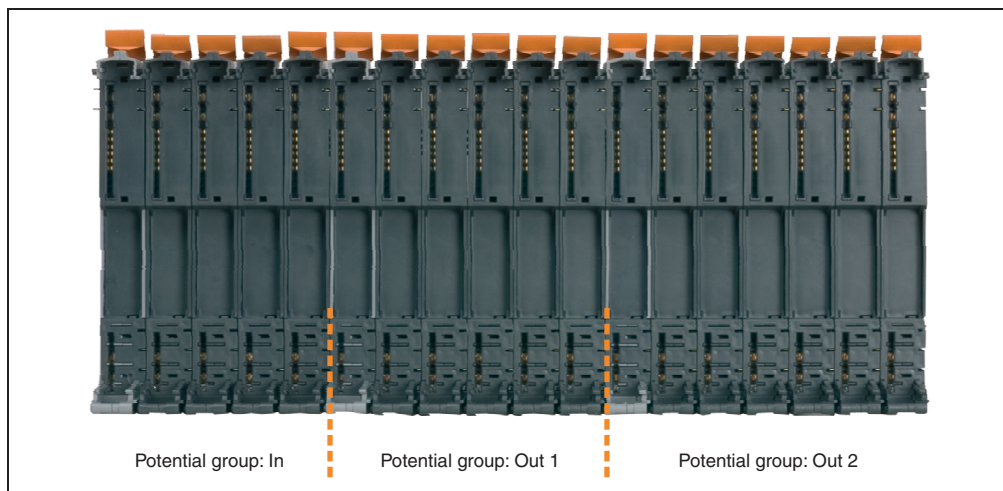


Figure 17: Simple implementation of various potential groups

Different potential groups can be implemented with the appropriate arrangement of supply bus modules, (e.g. for input groups or different E-stop circuits on the outputs.) The I/O supply is fed from supply modules.

8. Bus supply

The X2X Link supply must be connected in specific locations because the decentralized X2X Link backplane and the I/O electronics are completely electrically isolated. To start, the bus receiver takes on this task. Another supply must be connected after approximately 30 modules (for an exact calculation, see section 17 "Power output table" on page 60). A supply module must be used for the X2X Link. On the same module, a separate feed for the I/O supply can also be connected.

9. Potential groups

The I/O supply is connected via the bus modules. The feed is connected via corresponding supply modules. This makes it possible to implement simple potential groups (e.g. for input groups or various output groups). Additionally, the corresponding bus module necessary for the separation of the internal I/O supply is also required for this separation.

10. Output modules with supply

Generally, a supply module is also necessary for current output modules with many channels such as the 8 channel output module with 2 amp outputs. This is not the case with the X20 System. With this module, the supply feed is directly on the module, thereby saving supply modules and construction width.

11. Bus receiver with supply

The BR9300 bus receiver for the X20 System is equipped with a feed for the X2X Link as well as the internal I/O supply. This way, no additional supply module is needed.

12. Supply module for internal I/O supply

The first I/O modules of an X20 System are supplied by the bus receiver. The internal I/O supply is refreshed via the PS2100 supply module.

13. Supply module for internal I/O supply and bus supply

The X2X Link is fed by the BR9300 bus receiver. After approx. 30 modules (for an exact calculation, see section 17 "Power output table" on page 60), an additional supply must be connected. The PS3300 supply module is used for this purpose. This module is equipped with a feed for the X2X link as well as the internal I/O supply.

14. Bus sender with supply

The BT9100 bus transmitter has an integrated I/O supply feed. This saves a supply module for the last potential group.

15. X20 System protection

The X20 system is protected according to the power supply design.

15.1 Potential groups

Different potential groups can be implemented using the BM01 bus module, and with the appropriate arrangement of supply bus modules, (e.g. for input groups or different E-stop circuits on the outputs).

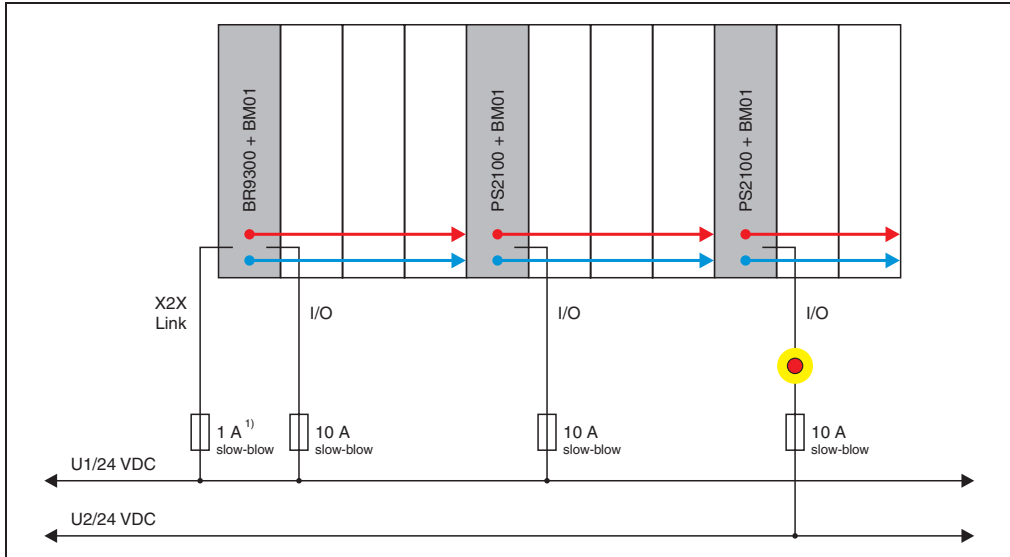


Figure 18: Protection of various potential groups

1) Recommended for line protection.

15.2 Supply feed via bus transmitter

The BT9100 bus transmitter has an integrated internal I/O supply feed. This saves a supply module for the last potential group.

Keep in mind: this potential group is separated from the rest of the potential groups by an I/O module with the BM01 bus module.

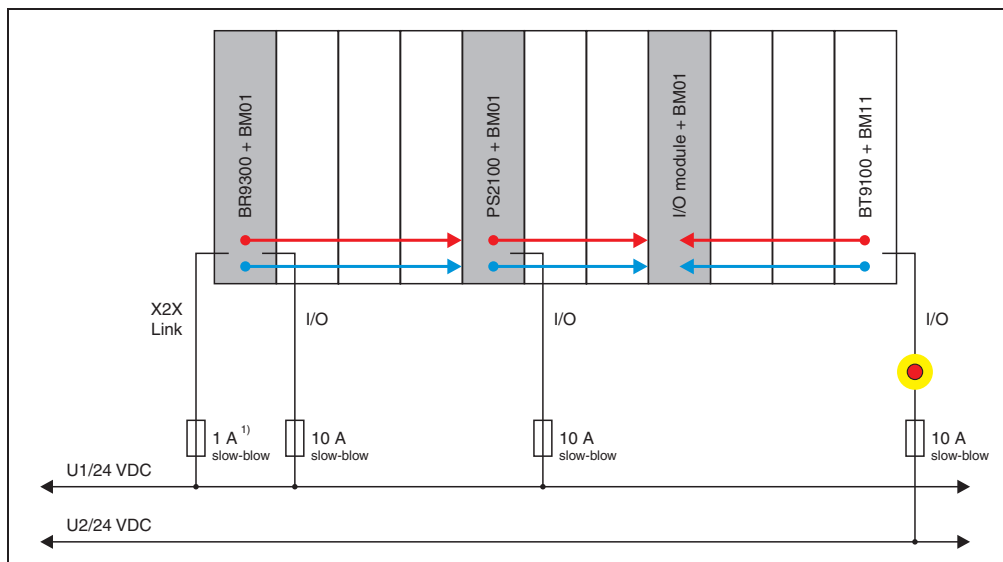


Figure 19: Protection for supply feed via bus transmitter

1) Recommended for line protection.

16. X2X Link supply

16.1 Expanded and redundant X2X Link supply

The remote backplane X2X Link is supplied separately from the I/O. This ensures that if there is a power outage on the I/O side (e.g. E-stop) the remote backplane will not be affected. After approx. 30 modules, a supply module for the X2X Link must be added.

To provide increased supply protection, it is possible to make the X2X Link supply redundant. To do this, at least one extra X2X supply module than would be needed to provide the required X2X Link performance must be used. This guarantees the function of the remote backplane even when one X2X Link supply goes down.

For proper calculation, note the following point:

- To determine the required X2X Link performance, calculate using 75 % of the supply module's rated power, during parallel operation.

Information:

If the X2X Link supply is not redundant or if the X2X Link supply of an X20 block is shut off completely, this should happen simultaneously for all supply modules.

16.2 Example of expanded X2X Link supply

Potential groups can be formed by the use of different types of supplies for the supply modules.

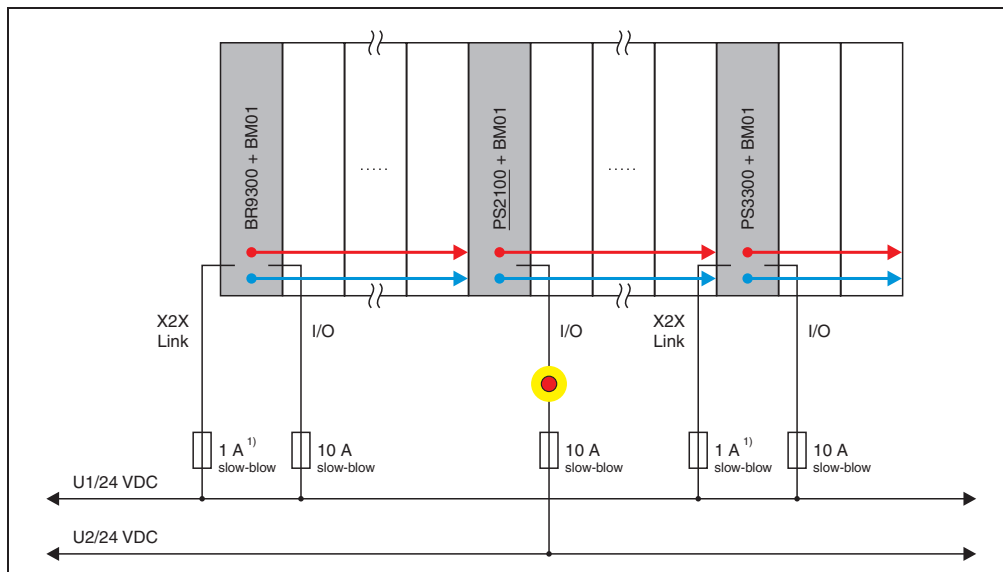


Figure 20: Example of expanded X2X Link supply

1) Recommended for line protection.

The PS3300 supply module supplies both the X2X Link and I/O, the PS2100 supply module only supplies the I/O.

16.3 Example of redundant X2X Link supply

Multiple PS3300 supply modules can be set up in parallel. Potential groups can be formed by the use of different types of supplies.

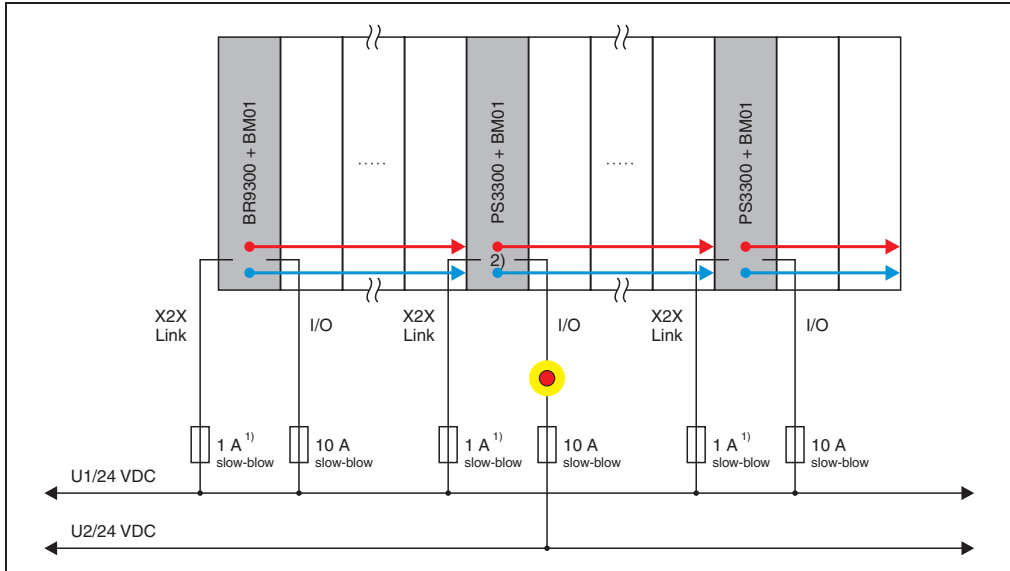


Figure 21: Example of redundant X2X Link supply

1) Recommended for line protection.

2) With split supplies, the two reference potentials (GND_1 and GND_2) are combined via the terminal block of the PS3300.

The PS3300 supply module supplies both X2X Link and the I/O.

17. Power output table

The "Bus power" and "Module I/O (internal)" columns contain values for the power provided by the module or the power required by the module. This allows a power output table to be calculated quickly and easily for a particular hardware configuration.

The value in the "Bus power" column refers to the power table for the X2X Link. The power supplied by the bus receivers and supply modules is labeled with a "+". The power required by a module is shown with a "-" sign. To calculate the power balance, the positive and negative power values should be added together. The sum may not be less than zero.

The values in the "Internal I/O power" column refer to the internal power requirements of the I/O modules using the 24 VDC I/O supply.

Product ID	Model number	Bus power [W] ¹⁾	Internal I/O power [W]
AI2622	X20AI2622	-0.01	-0.8
AI2632	X20AI2632	-0.01	-1.2
AI4622	X20AI4622	-0.01	-1.1
AI4632	X20AI4632	-0.01	-1.5
AO2622	X20AO2622	-0.01	-1.1
AO2632	X20AO2632	Rev. <B0: -0.01 Rev. ≥B0: -0.01	Rev. <B0: -1.6 Rev. ≥B0: -1.2
AO4622	X20AO4622	-0.01	-1.5
AO4632	X20AO4632	Rev. <B0: -0.01 Rev. ≥B0: -0.01	Rev. <B0: -2.0 Rev. ≥B0: -1.6
AT2222	X20AT2222	-0.01	-1.1
AT2402	X20AT2402	-0.01	-0.72
AT4222	X20AT4222	-0.01	-1.1
AT6402	X20AT6402	-0.01	-0.91
BB80	X20BB80	-	-
BC0043	X20BC0043	-1.5	-
BC0053	X20BC0053	-1.5	-
BC0063	X20BC0063	-2.3	-
BC0073	X20BC0073	-1.5	-
BC0083	X20BC0083	-2.0	-
BM01	X20BM01	-0.13	-
BM11	X20BM11	-0.13	-
BR9300	X20BR9300	+7.0	+240.0 ²⁾
BT9100	X20BT9100	-0.45	-
DC1196	X20DC1196	-0.01	-1.5
DC1198	X20DC1198	-0.01	-1.5

Table 1: X20 modules - overview for calculating the power output table